

Table 8 - Cross-family comparison with 95% bootstrap CI (d >= 1 km test subset)

Family	Model	RMSE (dB) [95% CI]	MAE (dB) [95% CI]	R2 [95% CI]	Bias (dB)
Uncalibrated empirical	ITU-R P.1546-6	13.64 [12.05, 15.15]	11.73 [10.06, 13.51]	-0.877 [-1.661, -0.410]	-8.87
Uncalibrated empirical	Hata standard	17.81 [14.64, 20.70]	13.76 [11.05, 16.66]	-2.201 [-3.511, -1.306]	13.54
Calibrated empirical	Hata L2 (OLS)	7.00 [5.58, 8.46]	5.37 [4.32, 6.59]	0.506 [0.256, 0.678]	-0.10
Pure ML	ML RF	4.78 [4.11, 5.45]	4.02 [3.41, 4.71]	0.770 [0.663, 0.835]	0.42
Hybrid A2	A2-CatBoost	4.74 [4.02, 5.46]	3.99 [3.38, 4.70]	0.773 [0.661, 0.843]	0.48